

الجامعة العربية الأمريكية
ARAB AMERICAN UNIVERSITY



FACULTY OF ENGINEERING

DEPARTMENT OF COMPUTER SYSTEMS ENGINEERING

Operating Systems

LINUX INSTALLATION

Dr. Sami Awad

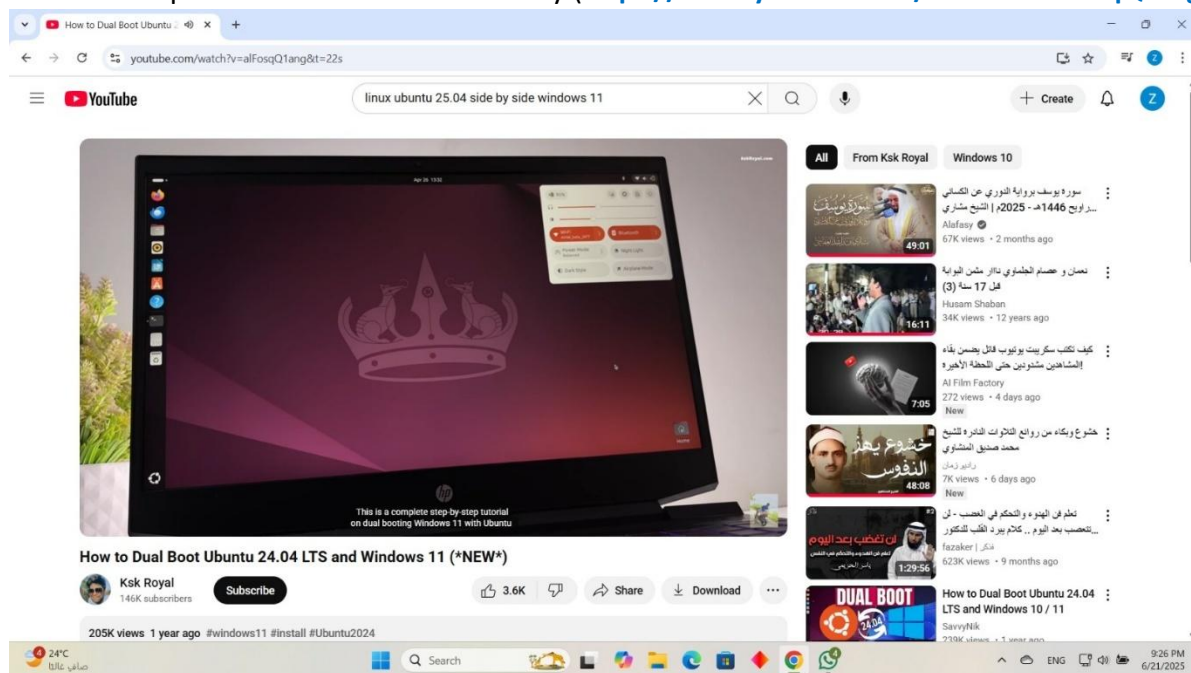
Zaid Abd Al Wahhab Ardah – 202210598

In this project, I installed the Linux-based operating system Ubuntu 25.04 side by side with Windows 11 in a dual-boot setup. I will walk through each step I followed during the installation process and share a brief overview of my experience using the system over two days. The goal of this project was to explore Ubuntu as an alternative operating system and to get hands-on experience with its features, interface, and performance compared to Windows.

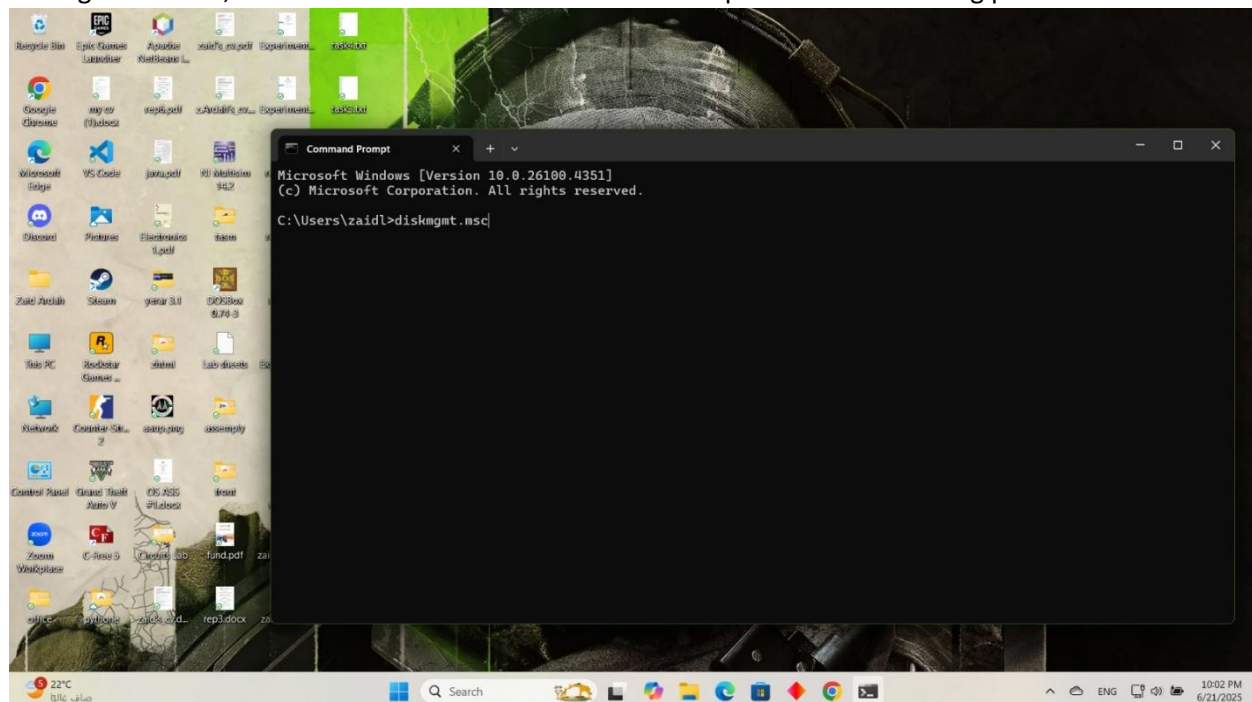
Before starting, I had Windows 11 installed on my device with its default graphical user interface (GUI), and it was the operating system I used regularly.

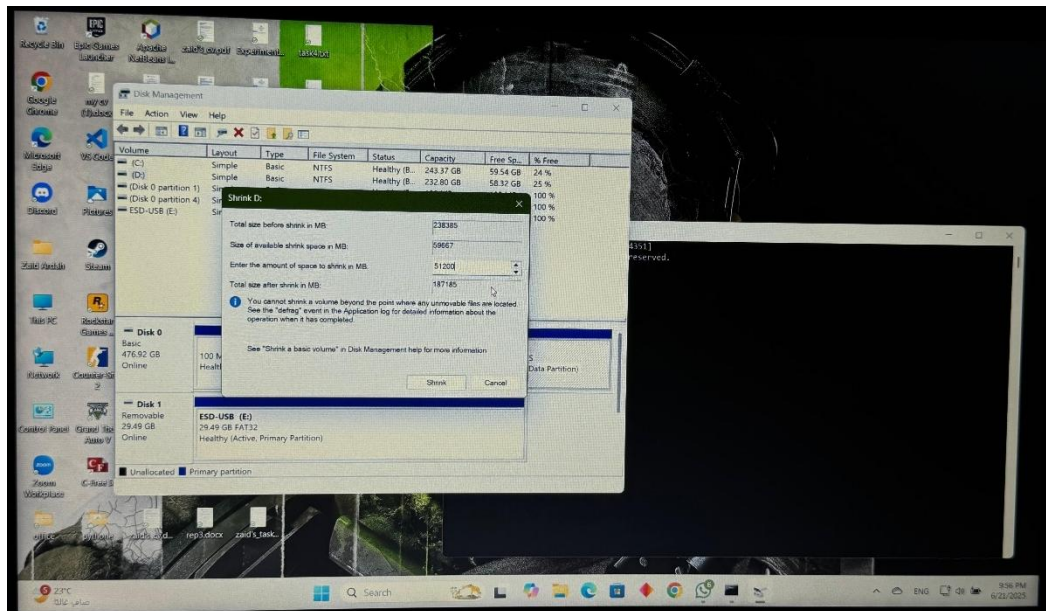
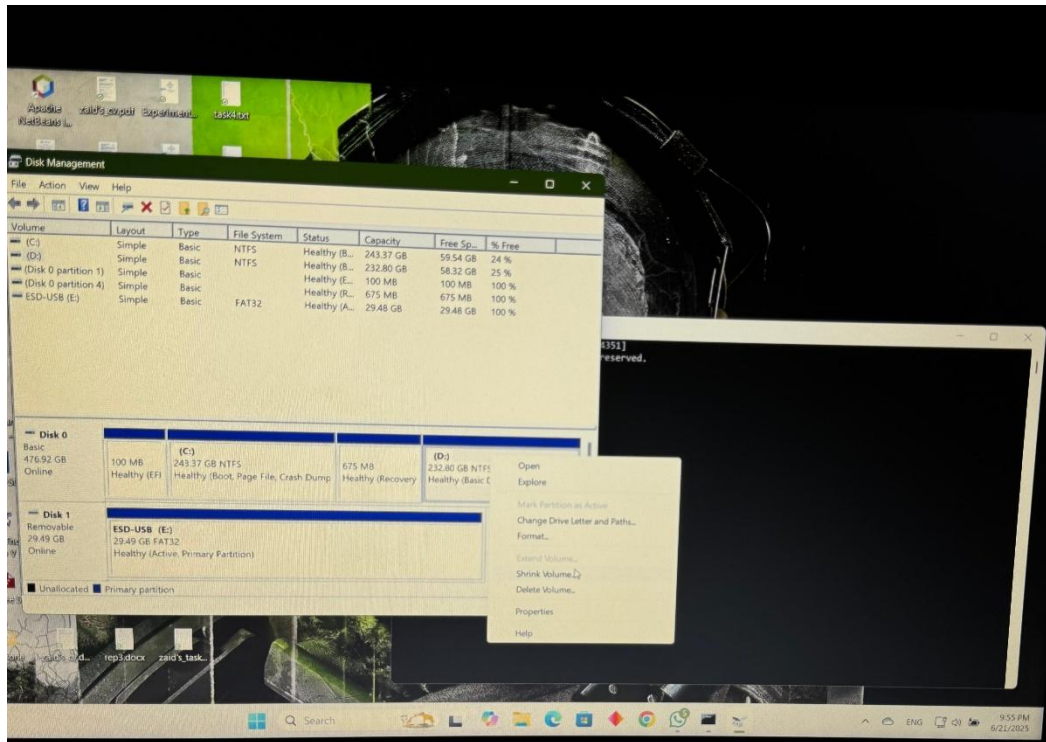


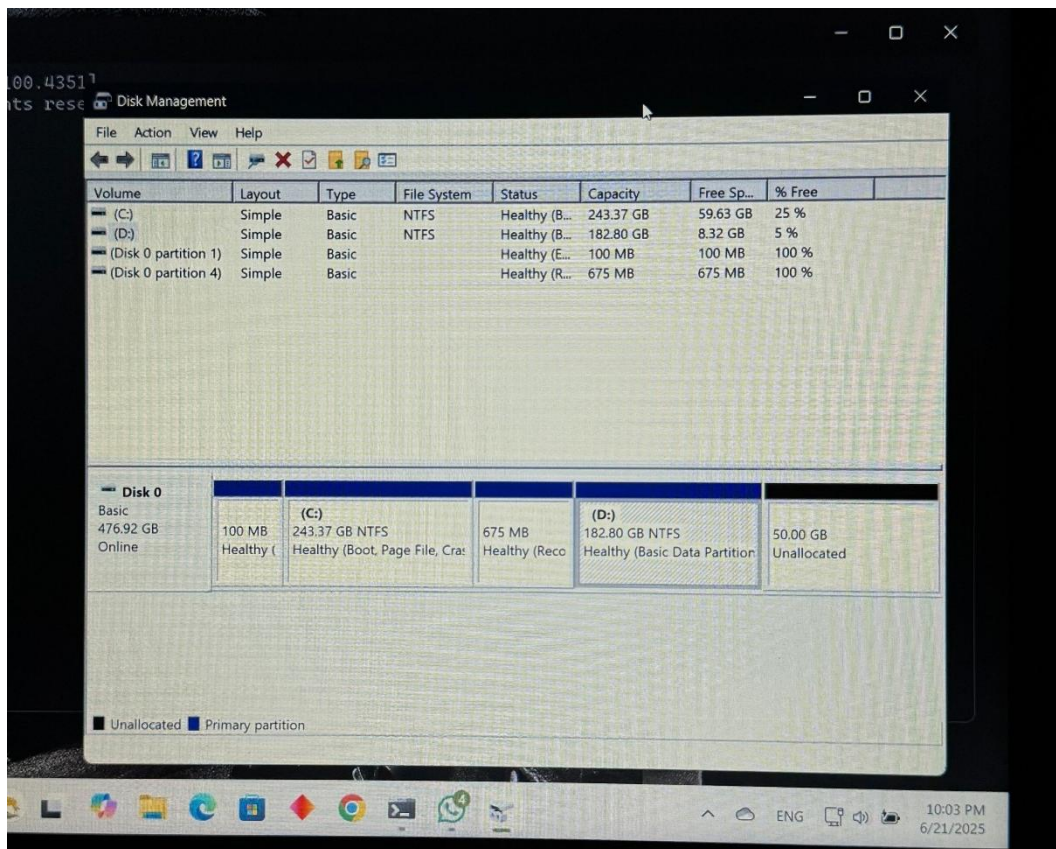
At the beginning, I watched a YouTube video (a screenshot of it is attached) that explained the installation process. After that, I started downloading Ubuntu and followed the steps and instructions provided in the video carefully (<https://www.youtube.com/watch?v=alFosqQ1ang>).



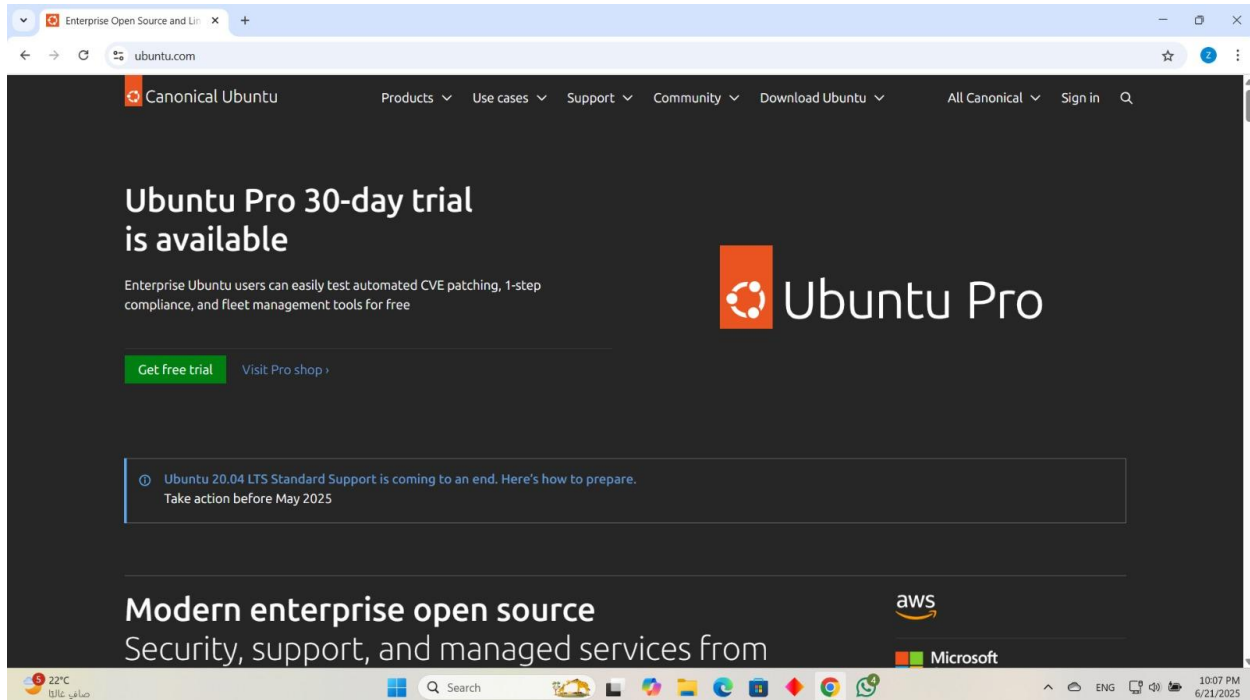
Then, I created unallocated space by shrinking the D drive and allocated 50 GB (51200MB) for the Linux installation. I typed the command **diskmgmt.msc** in the Command Prompt (**CMD**) to open the Disk Management tool, which allowed me to access the SSD and perform the shrinking process.







After that, I visited the official website ubuntu.com to download the latest version of Linux Ubuntu, which is 25.04.



Enterprise Open Source and Lin x +

ubuntu.com/#download-ubuntu

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Ubuntu 20.04 LTS Standard Support is coming to an end. Here's how to prepare.
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https://ubuntu.com/download/desktop

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Ubuntu 25.04



The latest version of the Ubuntu operating system for desktop PCs and laptops, Ubuntu 25.04 comes with nine months of security and maintenance updates, until January 2026.

Intel or AMD 64-bit architecture **Download** 5.9GB

ARM 64-bit architecture **Download** 3.6GB

For other versions of Ubuntu Desktop including torrents, the network installer, a list of local mirrors and past releases [check out our alternative downloads](#).

What's new System requirements How to install

- ✓ Linux Kernel 6.14 with support for the latest hardware
- ✓ GNOME 48 desktop environment with new Preserve battery mode and Wellbeing panel
- ✓ Papers as its default new PDF reader
- ✓ BeaconDB as its new geolocation provider
- ✓ Enhanced installer with improved encryption options and interaction with

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After that, I downloaded Rufus from its website. I used it to burn the Ubuntu ISO file onto the USB drive so it would be bootable for installation.

The screenshot displays a Windows 11 desktop environment. The top portion shows a web browser window with the Rufus website (rufus.ie/en/) open. The website provides information about the software, including a 'Download' button, a table of 'Latest releases', and 'System Requirements'. The bottom portion of the screenshot shows a Windows File Explorer window open to the 'Downloads' folder. The file list includes 'ubuntu-23.04-desktop-amd64.iso', 'rufus-4.9.exe', and several Word documents related to 'Project Linux Ubuntu Installation'. The taskbar at the bottom shows the system clock as 10:12 PM on 6/21/2025.

Thank you for downloading Rufus - Create bootable USB drive

rufus.ie/en/

- you need to work on a system that doesn't have an OS installed
- you need to flash a BIOS or other firmware from DOS
- you want to run a low-level utility

Despite its small size, Rufus provides everything you need!

A non exhaustive list of Rufus supported ISOs is also provided at the bottom of this page. [\(1\)](#)

Download

Latest releases:

Link	Type	Platform	Size	Date
rufus-4.9.exe	Standard	Windows x64	2 MB	2025.06.15
rufus-4.9p.exe	Portable	Windows x64	2 MB	2025.06.15
rufus-4.9_x86.exe	Standard	Windows x86	1.9 MB	2025.06.15
rufus-4.9_arm64.exe	Standard	Windows ARM64	6 MB	2025.06.15

[Other versions \(GitHub\)](#)
[Other versions \(FossHub\)](#)

System Requirements:
Windows 8 or later. Once downloaded, the application is ready to use.

Supported Languages:
Bahasa Indonesia, Bahasa Malaysia, Български, Čeština, Dansk, Deutsch, Ελληνικά, English, Español, Français, Hrvatski, Italiano, Latviešu, Lietuvių, Magyar, Nederlands, Norsk, Polski, Português, Português do Brasil, Русский, Română, Slovenščina, Srpski, Suomi, Svenska, Tiếng Việt, Türkçe, Українська, 简体中文, 正體中文, 日本語, 한국어, ไทย.

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Downloads

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New

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Details

Name	Date modified	Type	Size
Today			
ubuntu-23.04-desktop-amd64.iso	6/21/2025 10:16 PM	Disc Image File	6,131,368 ...
rufus-4.9.exe	6/21/2025 10:12 PM	Application	2,054 KB
Project Linux Ubuntu Installation (1).docx	6/21/2025 8:35 PM	Microsoft Word D...	16 KB
Earlier this week			
Project Linux Ubuntu Installation.docx	6/19/2025 8:13 PM	Microsoft Word D...	16 KB
C-FreePro-v5 (1).rar	6/15/2025 12:45 PM	WinRAR archive	22,894 KB
Earlier this month			
Operating Systems Syllabus ABET new v3...	6/5/2025 1:31 AM	Microsoft Word 97...	1,072 KB
Operating Systems Syllabus ABET new v3...	6/5/2025 12:56 AM	Microsoft Word 97...	1,072 KB
Last month			
Report Circuit Lab.rar	5/25/2025 11:56 PM	WinRAR archive	41,410 KB
Experiment#5 update (1).docx	5/25/2025 11:47 PM	Microsoft Word D...	211 KB
Experiment#4 update (1).docx	5/25/2025 11:47 PM	Microsoft Word D...	213 KB
Experiment#6 update.docx	5/25/2025 11:46 PM	Microsoft Word D...	195 KB
Experiment#5 update.docx	5/25/2025 11:46 PM	Microsoft Word D...	211 KB
Introduction.pdf	5/6/2025 12:01 AM	Chrome PDF Docu...	1,940 KB
Experiment#4 update.docx	5/4/2025 11:56 PM	Microsoft Word D...	213 KB
Lab Template - Electrical Engineering (1)...	5/4/2025 11:31 PM	Microsoft Word D...	587 KB
Experiment#3 update (1).docx	5/4/2025 11:31 PM	Microsoft Word D...	264 KB
AnyDesk.exe	5/3/2025 5:45 PM	Application	7,503 KB
Earlier this year			
Experiment#3 update (1).docx	4/28/2025 5:56 PM	Microsoft Word D...	764 KB

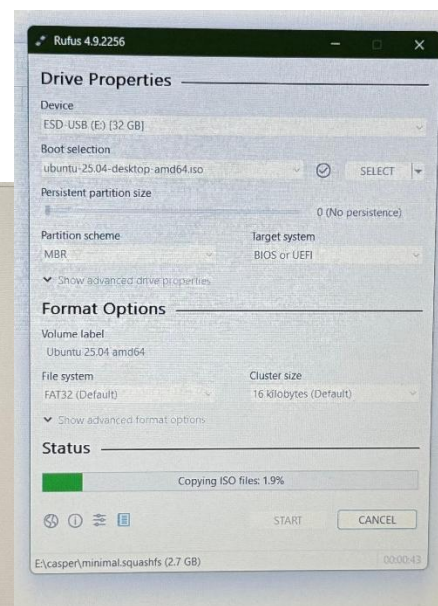
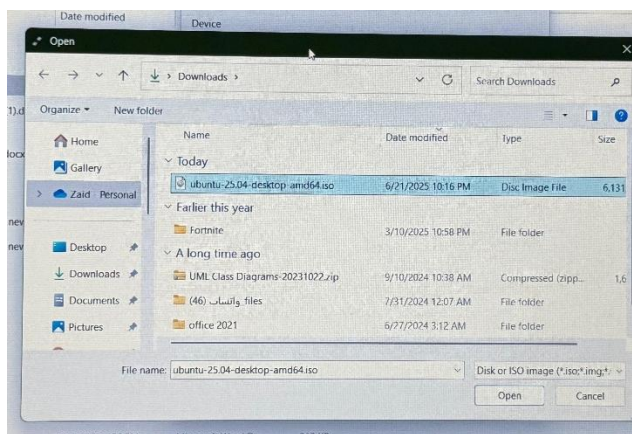
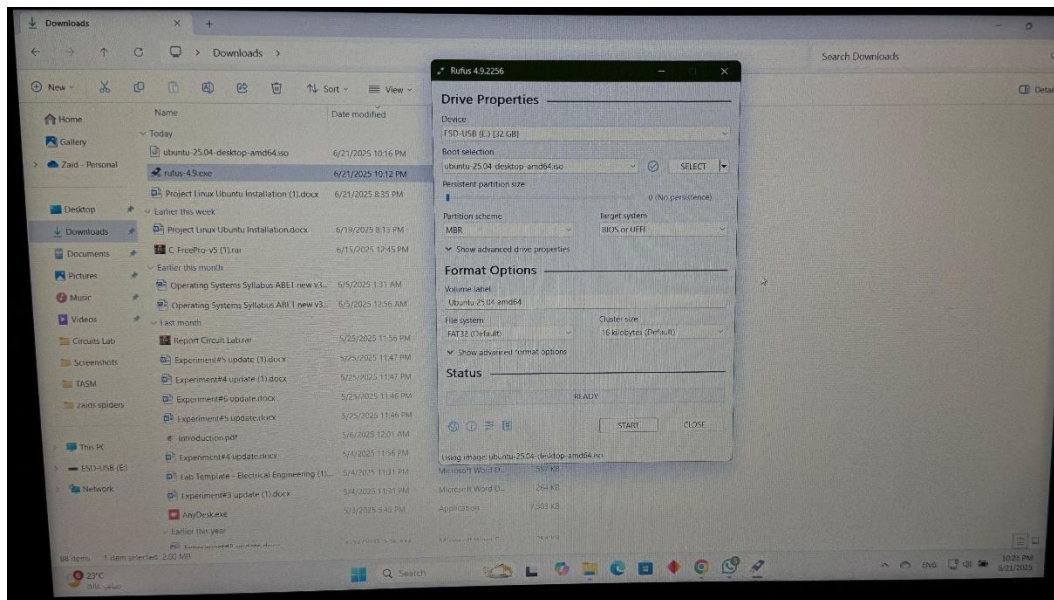
88 items

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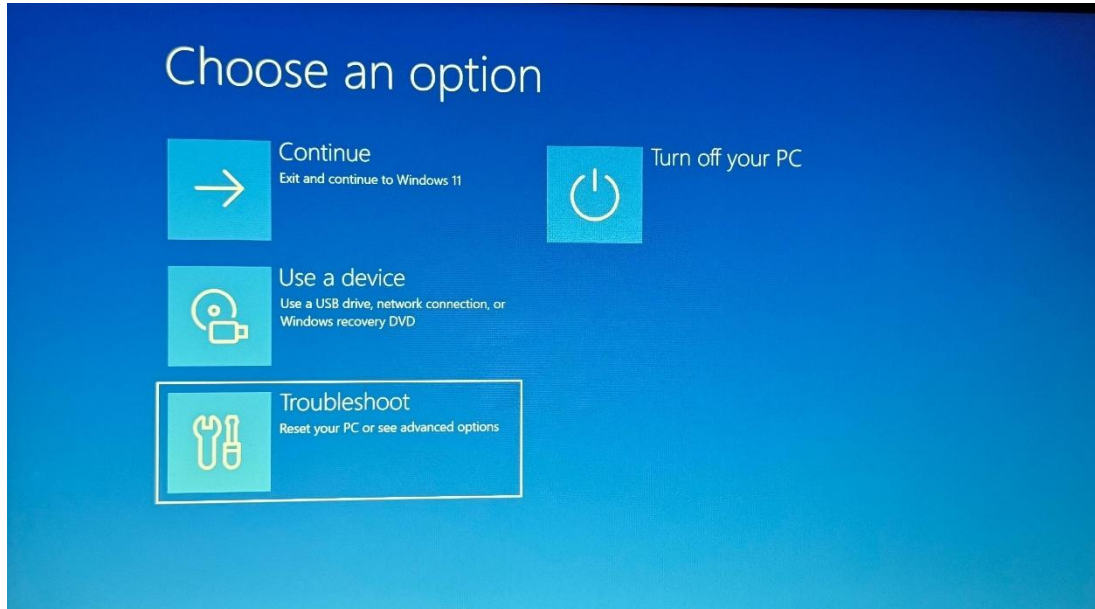
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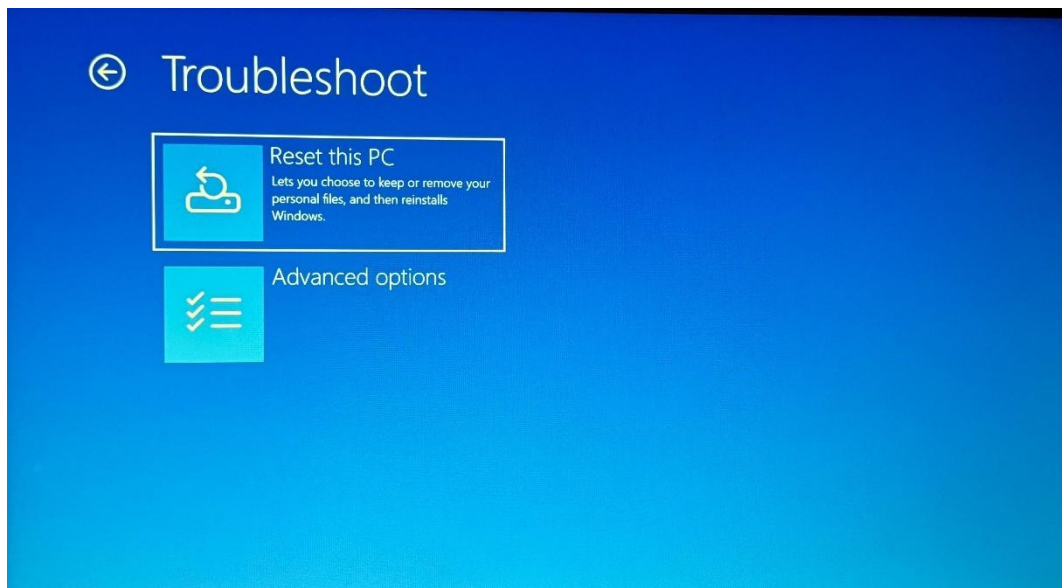
When I opened Rufus, I clicked to select the Linux ISO file that I had downloaded earlier. After choosing the file, I just left the default settings as they were, and Rufus was ready to create the bootable USB.



After that, I restarted the device and continued pressing the Shift key, and this window opened.



And from here, we started the actual installation, as I accessed Troubleshoot from here.



Then to Advanced Options.



From here, I clicked on **UEFI Firmware Settings**.

This option allows access to the computer's BIOS/UEFI interface, where I can change hardware-level settings, such as boot order, secure boot, and virtualization features. It is an essential step if you want to boot from a USB drive or make BIOS-level changes before installing or repairing Windows.

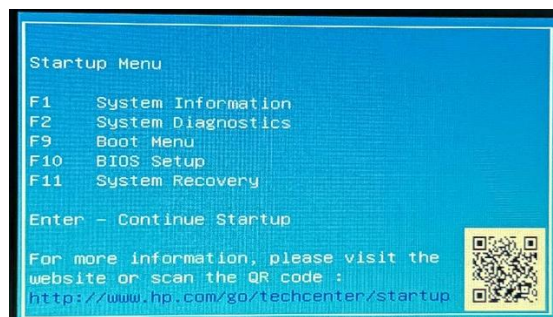
Once I clicked on **UEFI Firmware Settings**, I was directed to this screen.

Here, I pressed the **Restart** button to reboot the system and enter the BIOS/UEFI interface.

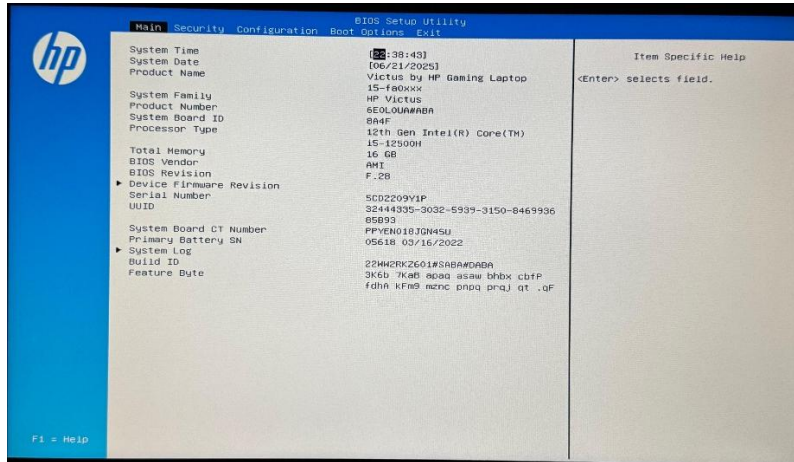


After restarting, I reached the **Startup Menu** screen. From here, I selected **F10: BIOS Setup** to enter the BIOS interface.

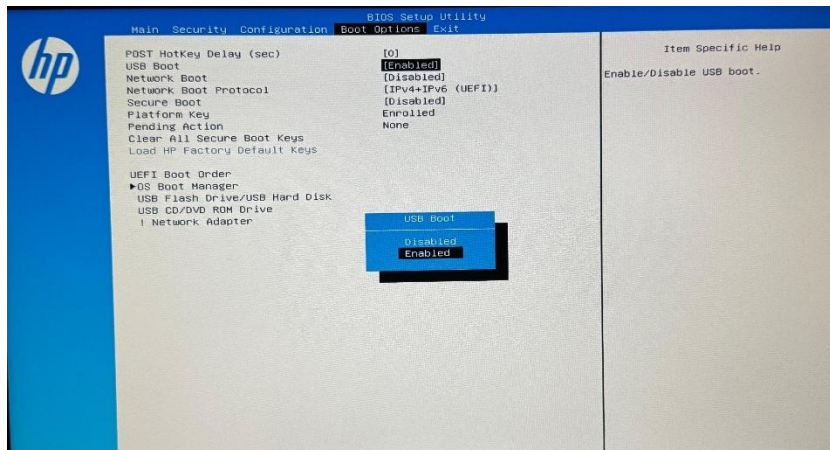
This step is crucial, as it allows me to configure hardware-level settings such as boot priority, enabling or disabling Secure Boot, and managing legacy/UEFI modes. These changes are often required before installing a new operating system or booting from external media like a USB drive.



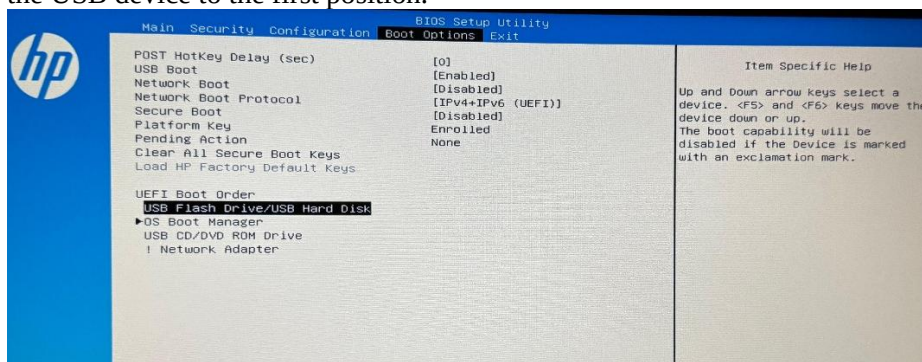
After that, I went to Boot Options.



Inside the **Boot Options** menu, I enabled the **USB Boot** feature. This is necessary to allow the system to boot from a USB drive, which I prepared with a Linux distribution. Enabling this option lets me proceed with installing Linux from the bootable USB.

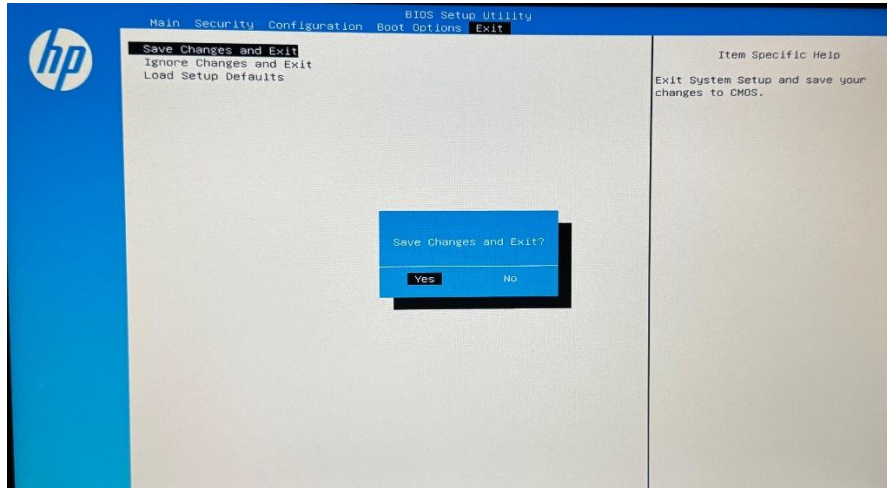


I changed the **UEFI Boot Order** by moving the **USB Flash Drive/USB Hard Disk** to the top of the list. This ensures that the system will boot from the USB drive first which contains the Linux installation media instead of booting into the currently installed operating system. I used the **F5** and **F6** keys to move the USB device to the first position.

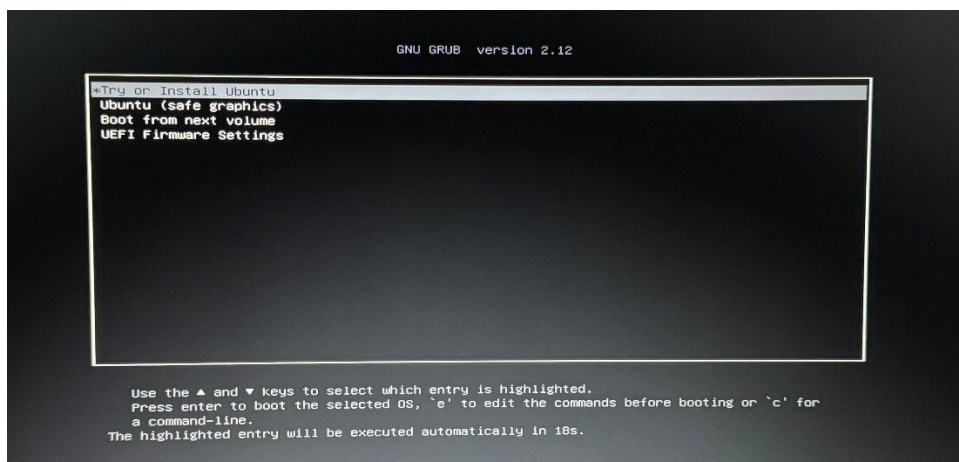


After setting the USB drive as the first boot device, I navigated to the **Exit** tab and selected Save Changes and Exit.

A confirmation message appeared, and I chose **Yes** to apply the changes. This saved my configuration and restarted the system, allowing it to boot directly from the Linux USB installer.



After restarting, the system successfully booted from the USB drive, and I was presented with the **GRUB menu**. Here, I selected "**Try or Install Ubuntu**" to proceed with launching the Ubuntu live environment, where I can either test Ubuntu or start the installation process on the internal drive.

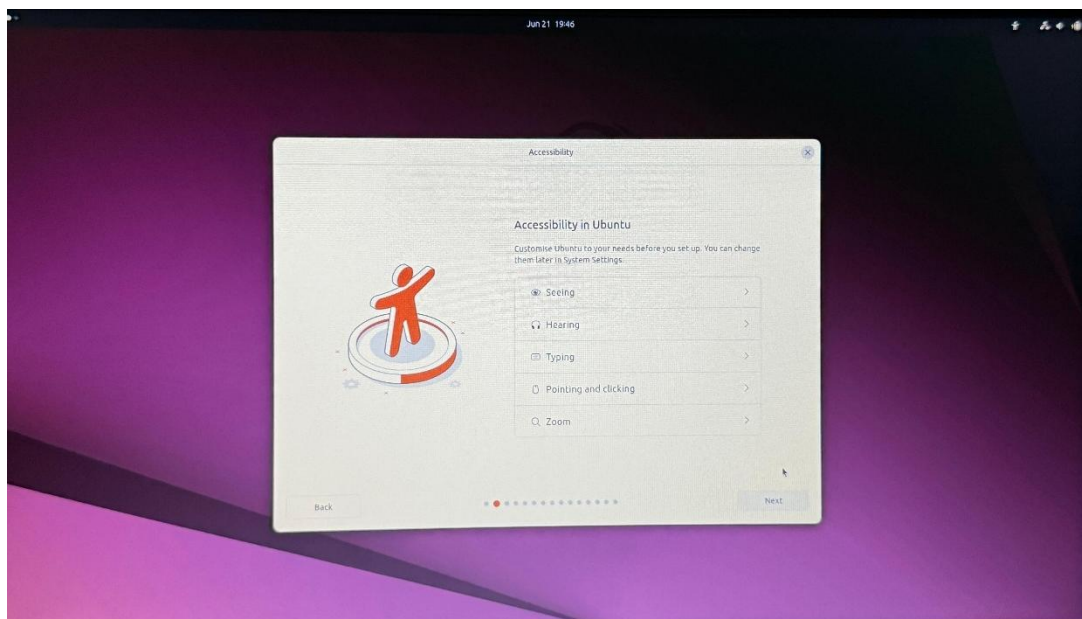
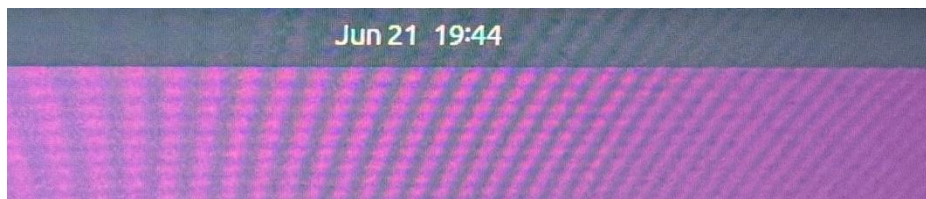
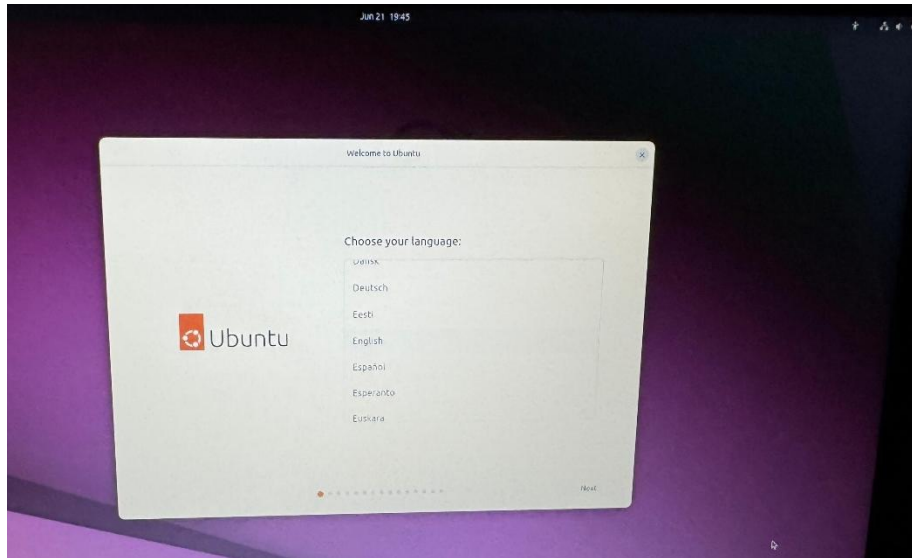


the system began booting into the Ubuntu live environment.

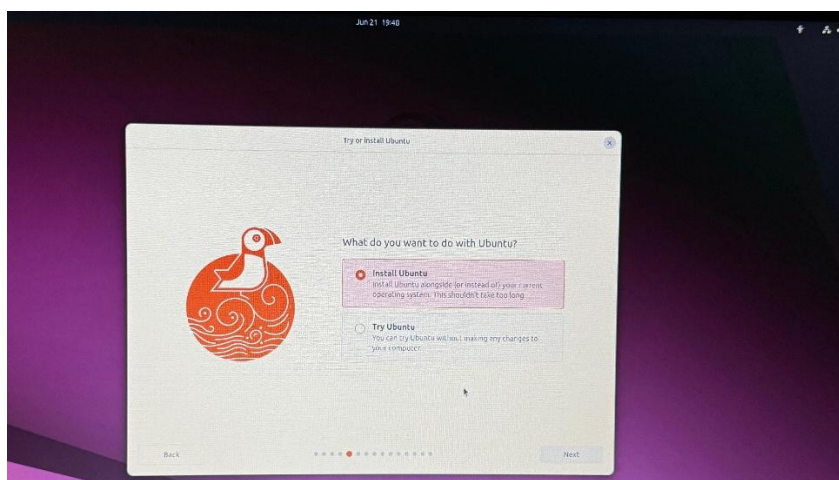
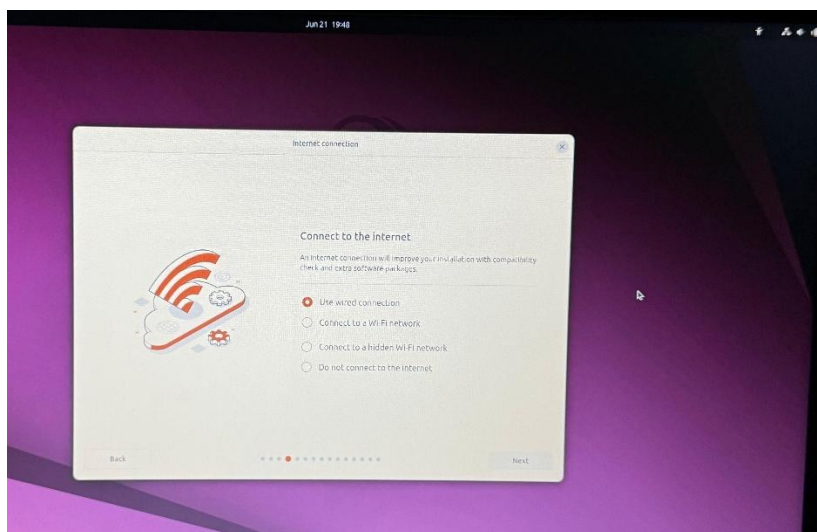
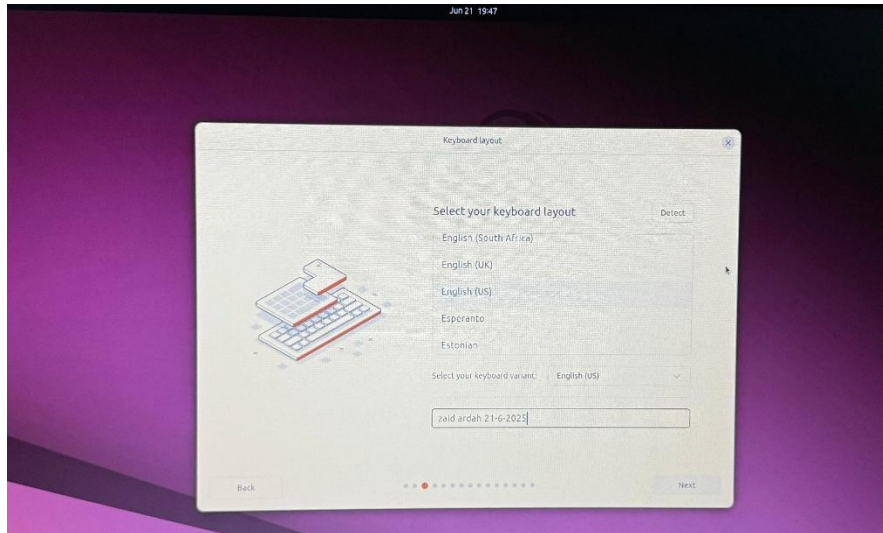
At this point, the Ubuntu logo appeared on the screen, indicating that the system is loading the necessary files from the USB to launch the installer interface.

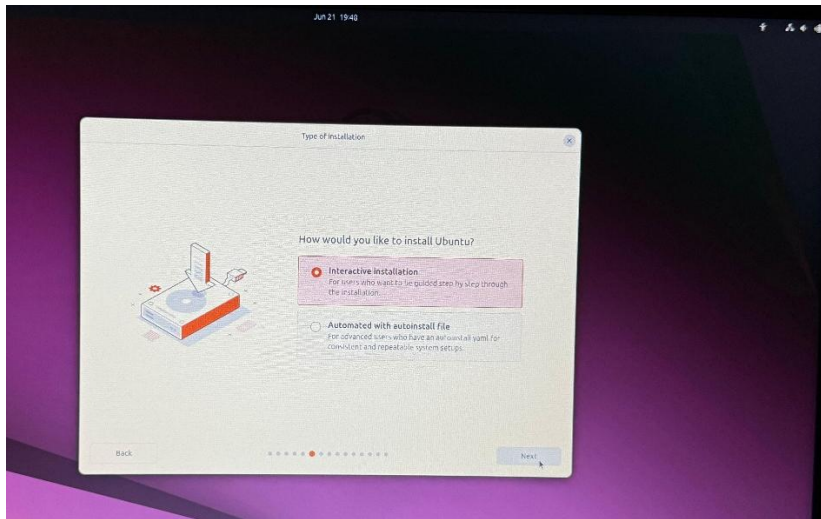


Now the Linux setup process has started, including language selection and other initial configuration steps.

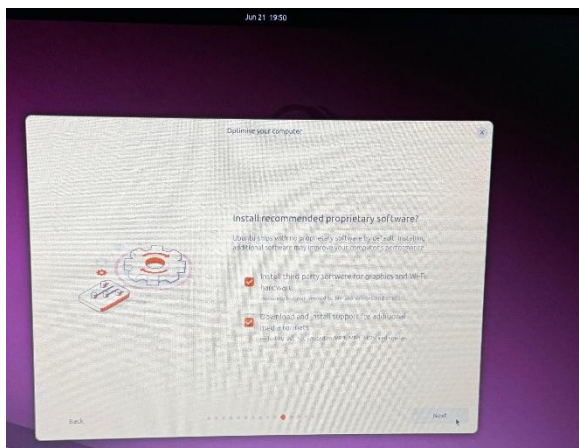
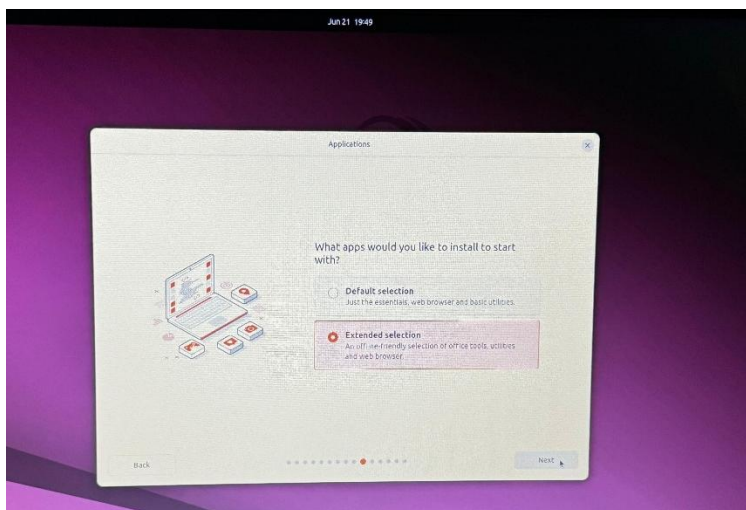


At this stage, I selected the keyboard layout and tested it by typing my name and the current date ("zaid arдах 21-6-2025") in the input field.

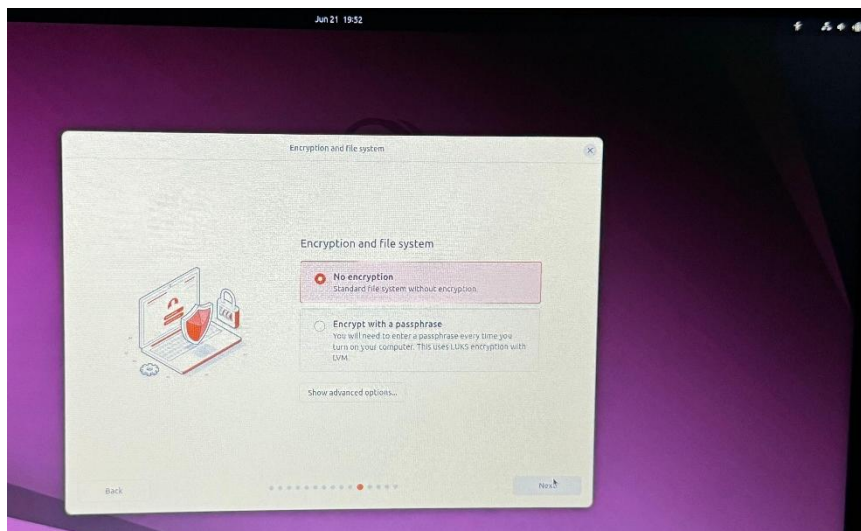
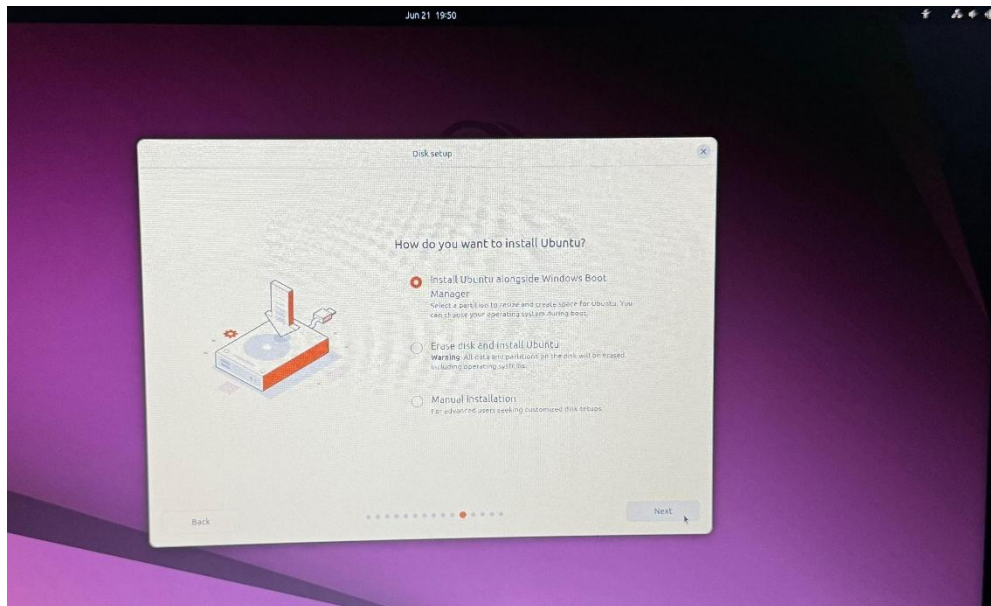




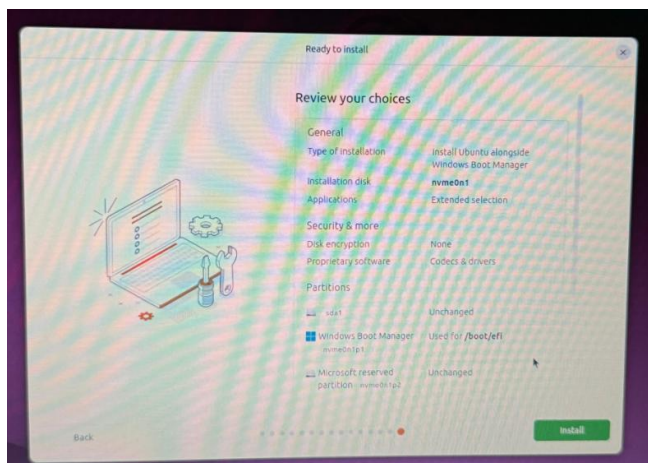
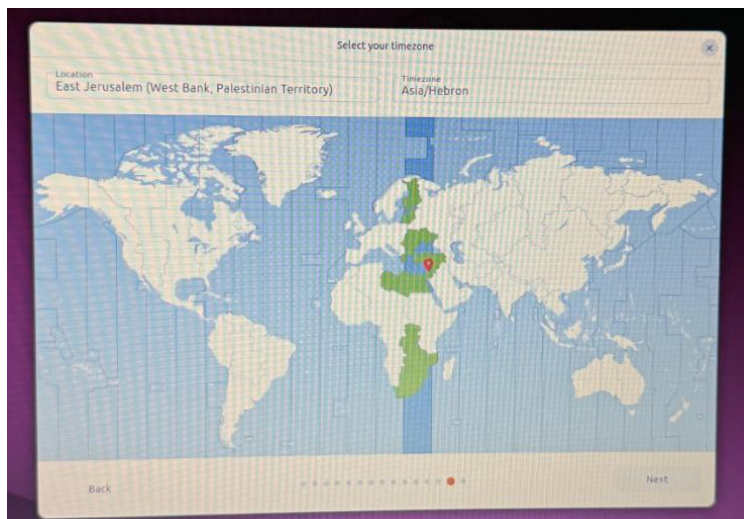
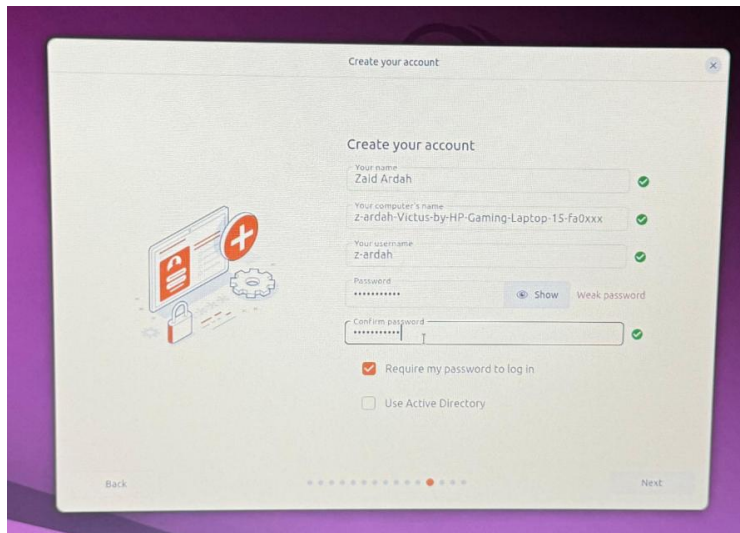
In this step, I was prompted to choose which set of applications I would like to install. I selected "**Extended selection**", which includes a more complete set of tools such as office software, utilities, and a web browser providing a ready-to-use environment right after installation.



In the disk setup screen, I selected the option “Install Ubuntu alongside Windows Boot Manager”. This allows Ubuntu to be installed side-by-side with the existing Windows operating system, creating a **dual-boot setup**. Upon system startup, I will be able to choose whether to boot into Ubuntu or Windows.

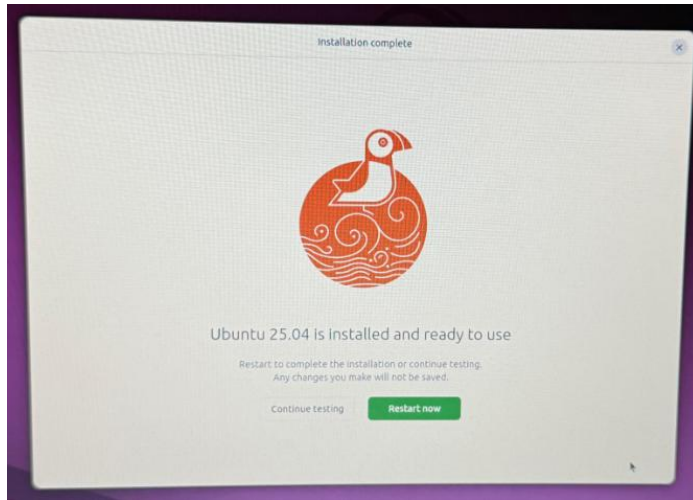


In this step, I created my user account by entering my name, username, and password. This account will be used to log in and manage the Ubuntu system. I also enabled the option to require a password to log in, adding a layer of security to the system.



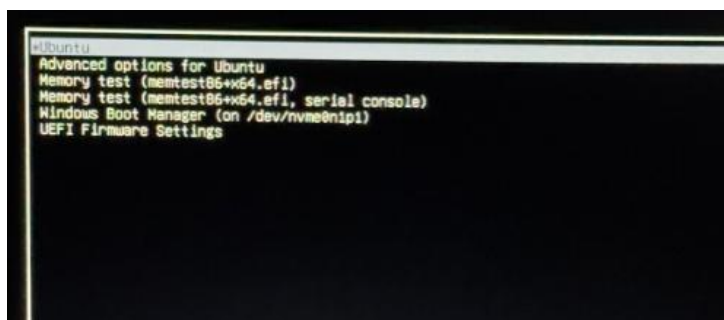
The installation process is now about to begin.

All settings have been confirmed, and Ubuntu will start installing alongside the existing system.

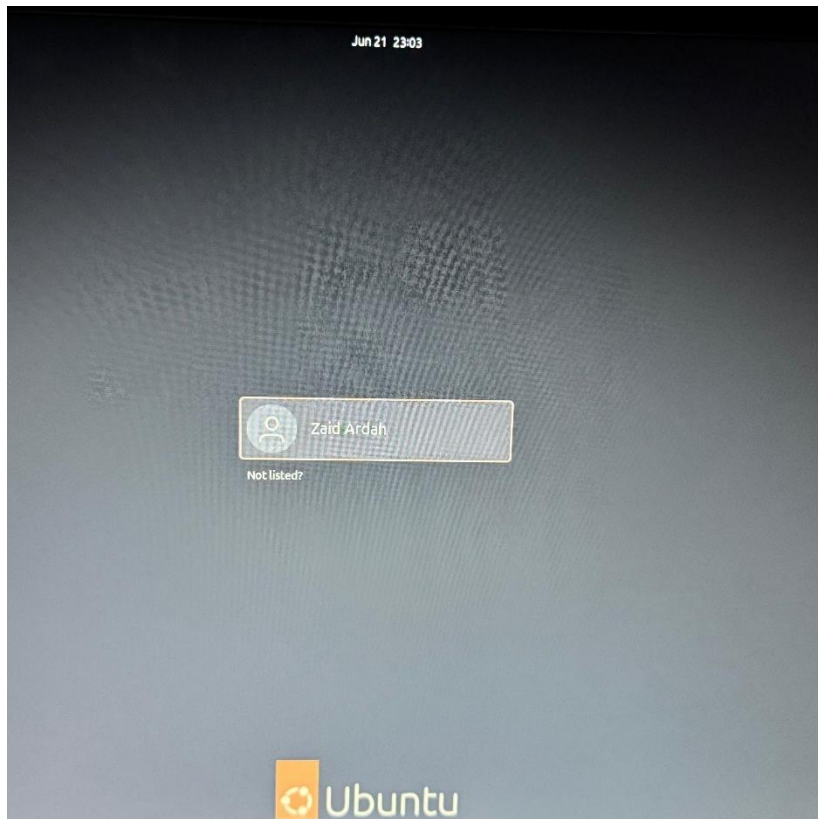


The installation has finished successfully.

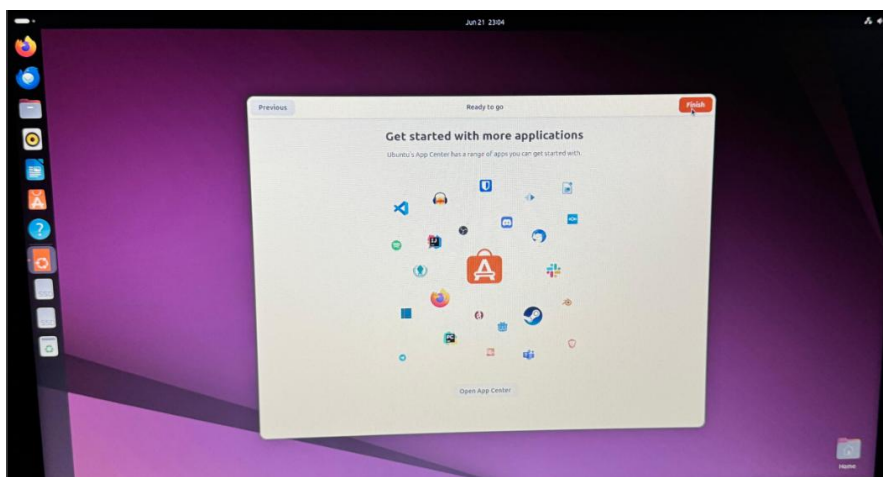
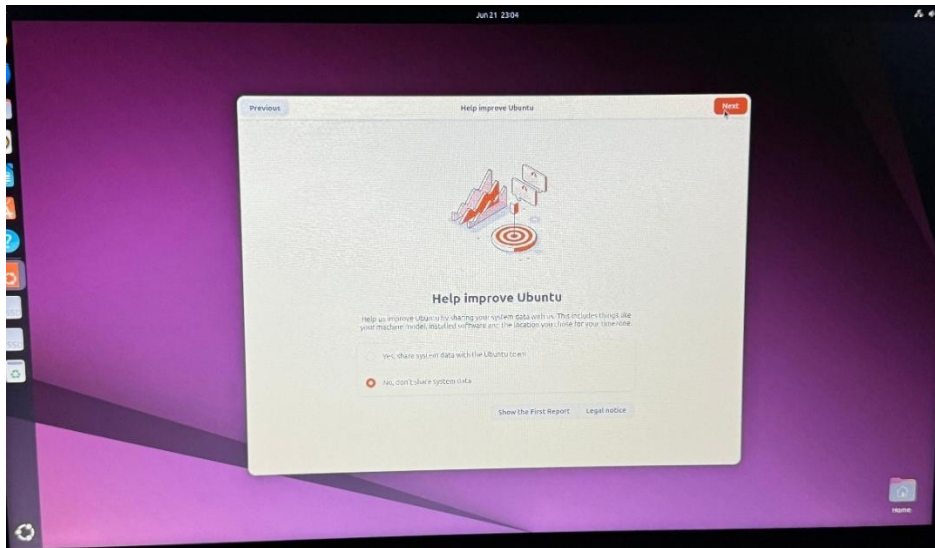
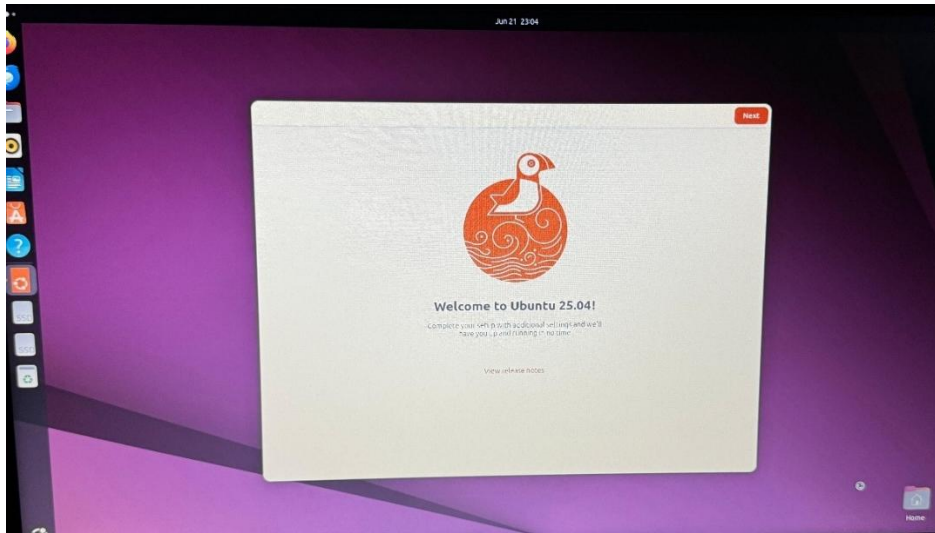
The system now prompts to **remove the installation medium (USB)** and **press ENTER** to reboot into the newly installed Ubuntu system. **After the restart, the system displayed a boot menu allowing me to choose which operating system to launch — either Ubuntu nor Windows.**



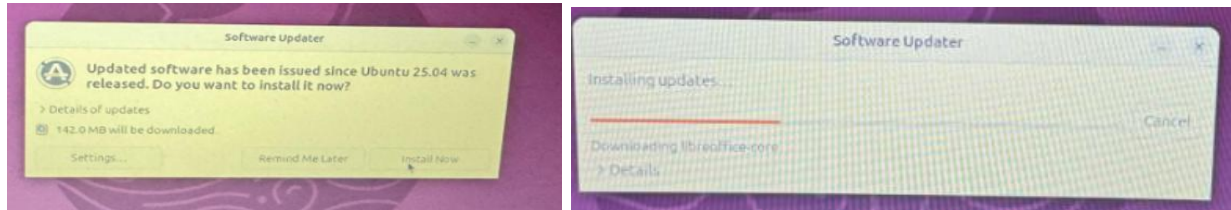
After selecting **Ubuntu** from the boot menu, the system began to load.
Once it reached the login screen, I entered my **user password** to access my personal account.
The Ubuntu desktop then loaded successfully, marking the first full login into the newly installed system.



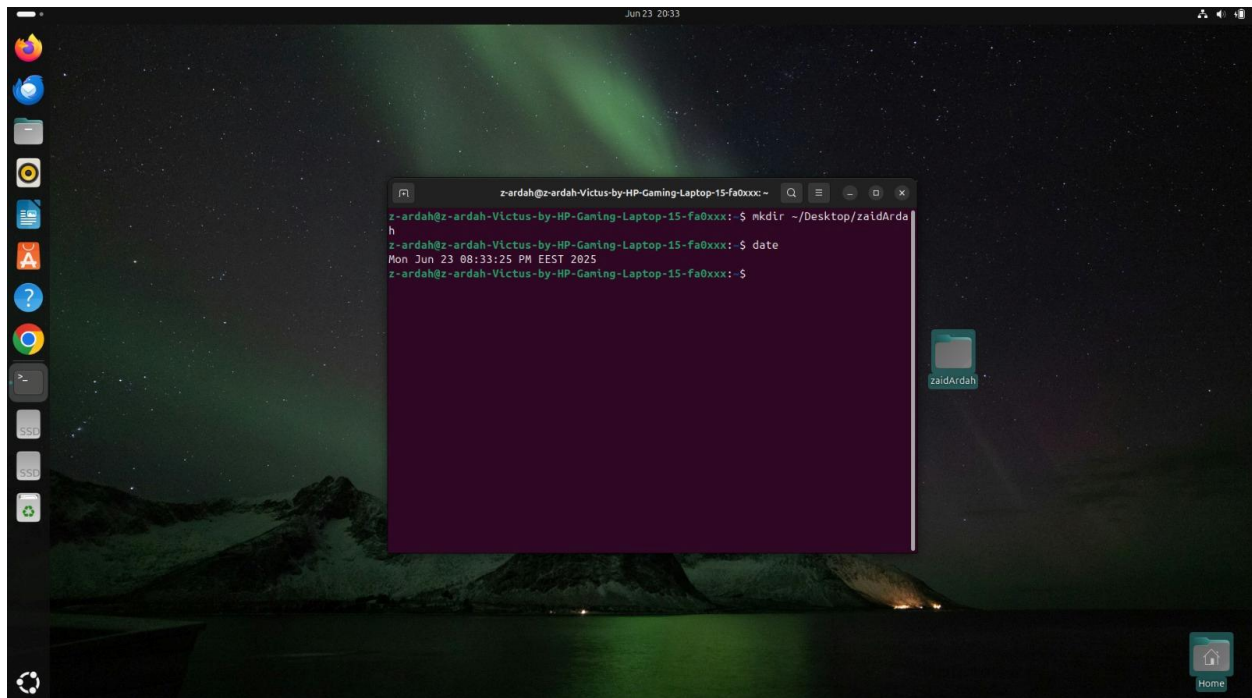
Finish the installation process:



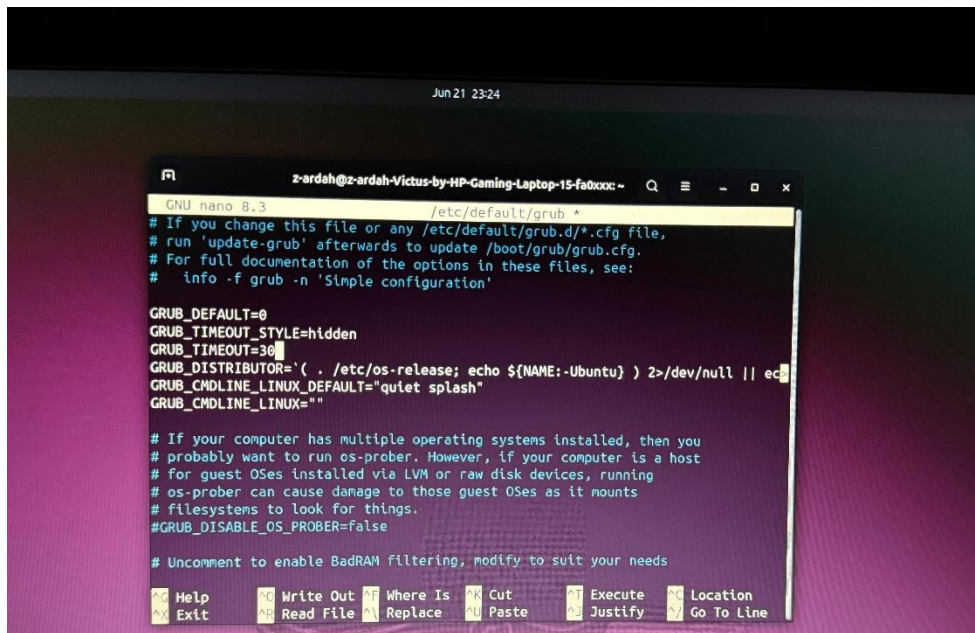
Shortly after logging into Ubuntu, I was prompted with a Software Updater message indicating that updates have been released since version Ubuntu 25.04. The system offered to download and install 142.0 MB of updates. I clicked "Install Now" to ensure the system is up to date and secure.



I opened the terminal on my Linux operating system and created a new folder on the Desktop named **zaidArdah** using the `mkdir` command. After creating the folder, I used the `date` command to confirm the current time, which was **08:33:25 PM**. This simple exercise helped me practice basic terminal commands and understand how to interact with the file system in a Linux environment.



Now, I want to show things from my own experience, I changed the GRUB boot menu timeout to 30 seconds. This gave me more time to choose between operating systems during startup, especially useful in a dual-boot setup with Ubuntu and Windows.



```
z-ardah@z-ardah-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~
GNU nano 8.3 /etc/default/grub
# If you change this file or any /etc/default/grub.d/*.cfg file,
# run 'update-grub' afterwards to update /boot/grub/grub.cfg.
# For full documentation of the options in these files, see:
#   info -f grub -n 'Simple configuration'

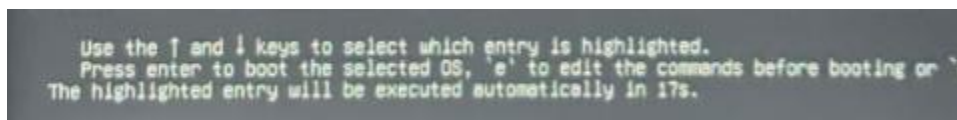
GRUB_DEFAULT=0
GRUB_TIMEOUT_STYLE=hidden
GRUB_TIMEOUT=30
GRUB_DISTRIBUTOR='(. /etc/os-release; echo ${NAME:-Ubuntu} ) 2>/dev/null || echo `lsb_release -d -s`'
GRUB_CMDLINE_LINUX_DEFAULT='quiet splash'
GRUB_CMDLINE_LINUX=""

# If your computer has multiple operating systems installed, then you
# probably want to run os-prober. However, if your computer is a host
# for guest OSes installed via LVM or raw disk devices, running
# os-prober can cause damage to those guest OSes as it mounts
# filesystems to look for things.
#GRUB_DISABLE_OS_PROBER=false

# Uncomment to enable BadRAM filtering, modify to suit your needs

^C Help      ^O Write Out ^T Where Is  ^K Cut       ^J Execute   ^G Location
^X Exit      ^R Read File ^I Replace  ^U Paste     ^_ Justify   ^Y Go To Line
```

One of the most surprising and unfamiliar things for me was when I entered my password after running the command: **"sudo nano /etc/default/grub"**. **Nothing** appeared on the screen as I typed the password - no characters, no asterisks, nothing at all. At first, I thought something was wrong or not working. But after doing some research, I found out that this is completely normal in Linux: the system does not display any visual feedback while typing passwords in the terminal, for security reasons.



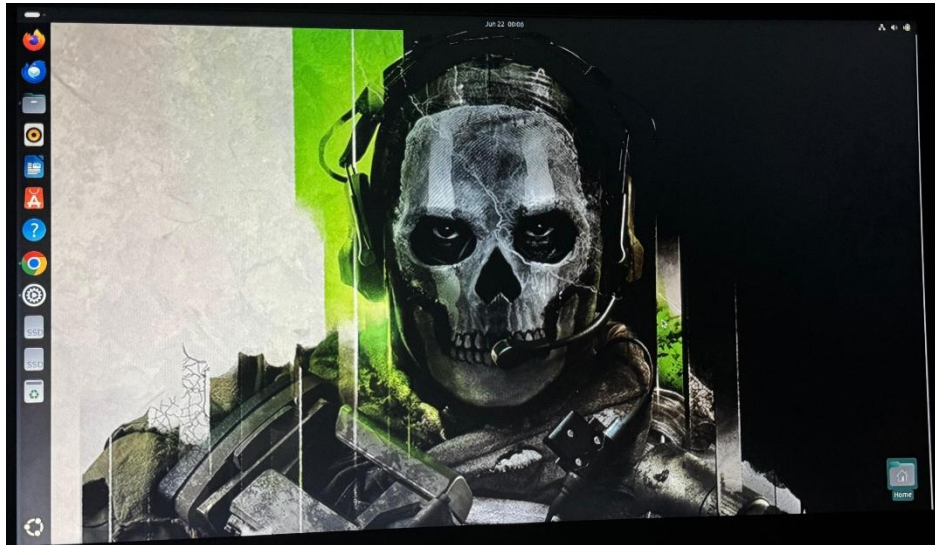
After that, I used the Terminal to install Google Chrome.

I downloaded the .deb package from the official website and installed it using terminal commands. This showed me how powerful and flexible the Linux Terminal is — many tasks can be done quickly and efficiently through it I also customized my desktop by changing the wallpaper.



**Finally, I set my favorite wallpaper on the Linux system, the same one I used on Windows.
At this point, I started to feel like I was getting used to the system and learning how to use it more comfortably.**

Each step helped me understand Linux better and made the experience less intimidating.



In conclusion, at the end of the Operating Systems course and after successfully installing the new Linux operating system,

I would like to express my deep gratitude and sincere thanks to you, Dr. Sami Awad, for everything you have provided us during the course, from the valuable knowledge to your clear and dedicated explanations.

Thank you for giving us this great opportunity at the end of the course to work hands-on with a new operating system.

This practical experience helped me apply what I learned and truly deepened my understanding of the course material.

Thank you very much, Doctor, for all your efforts.

I truly hope I was a dedicated and hardworking student.

Wishing you all the best, and I hope to see you again in future courses, with full appreciation and respect.

With sincere gratitude,

Your student, Zaid Ardah

23/6/2025.